Fine-Scale Urban Informal Settlements Mapping by Fusing Remote Sensing Images and Building Data via a Transformer-Based Multimodal Fusion Network

Runyu Fan, Fengpeng Li, Wei Han, Jining Yan, *Member, IEEE*, Jun Li, *Fellow, IEEE*, and Lizhe Wang, *Fellow, IEEE*

***Abstract*— Urban informal settlements (UISs) are high-density population settlements with low standards of living and supply. UIS semantic segmentation, which identifies pixels corresponding to informal settlements in remote sensing images, is crucial to the estimation of poor communities, urban management, resource allocation, and future planning, particularly in megacities. However, most studies on informal settlement mapping are either based on parcels (image classification) or pixels (semantic segmentation). Few studies utilize object information to improve UIS mapping. Since informal settlements are formed by buildings (objects), utilizing object information can improve UIS semantic segmentation. Furthermore, current UIS mapping studies mainly focus on using single-modality remote sensing images, and there is a lack of related research on using multimodal data. Due to the spatial heterogeneity of informal settlements, using only a single modality of remote sensing image features limits the effectiveness and accuracy of informal settlements semantic segmentation. Aiming at achieving fine-scale UIS mapping results, this article proposes a UIS semantic segmentation method, namely UisNet, that utilizes a transformer-based block to receive multimodal data, including high-spatial-resolution remote sensing images (parcel- and pixel-level) and building polygon data (object-level) to identify UIS. The experiments were conducted in Shenzhen City, and they confirmed the superior performance of UisNet, which achieved an overall accuracy (OA) of 94.80% and a mean intersection over union (mIoU) of 85.51% in the testing set of the manually labeled UIS semantic segmentation dataset (UIS-Shenzhen dataset) and outperformed the best models on semantic segmentation tasks. Besides, we add a set of experiments on a public dataset [gaofen image dataset (GID) dataset] and compare our method with the current state-of-the-art semantic segmentation methods. Experiments show that the proposed UisNet improves mIoU by 1.64% to 7.58% compared to other methods. This work will be available at https://github.com/RunyuFan/.**

***Index Terms*— Deep learning, multimodal, remote sensing, semantic segmentation, urban informal settlements (UISs).**

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I. Introduction

URBAN informal settlements (UISs) are high-density population settlements characterized by durable housing, low living standards, poor access to safe water, and poverty. Today, urban areas are home to many human activities, social functions, and more than half the population. Urban development contains quantitative expansion and qualitative improvement. Quantitative expansion means an increase in the number and scale of cities, that is, an increase in urbanization. Qualitative improvement means the development of urban functions and modernization. With the ongoing quantitative expansion and qualitative urban development, there has been an increase in the number and scale of cities, and various megacities have witnessed the strengthening of urban functions and modernization levels. However, some UIS are hampered by traffic congestion, few public facilities, and limited medical resources. Detailed information on the location and boundaries of informal settlements is of great help for social-economic studies and, thus, provides beneficial insights for sustainable urban planning. UIS mapping aims to automate the recognition of informal settlements in cities, and it plays a significant role in city's sustainable development, population estimation, and future planning.

With the launch of various satellites in recent years, remote sensing images can now be acquired with a higher spatial resolution, making it possible to grasp the physical spatial settlement structures and the visual characteristics of UIS. Higher resolution remote sensing images and more diverse geospatial data from Internet of Things (IoT) devices in cities have encouraged many machine learning-based (ML-based) approaches to identifying UIS in urban areas. Therefore, increased interest in employing high-resolution remote sensing images and big geospatial data in the UIS mapping topic has been noted [1], [2], [3]. Inspired by the progress made by ML-based models in various urban computing and remote sensing image interpreting tasks, including urban functional zones mapping [4], [5], [6], [7], remote sensing scene classification [8], [9], [10], land use and land cover mapping [11],

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[12], [13], remote sensing object detection [14], [15], [16], etc., more and more studies have broadened these highly effective models for dealing with UIS mapping tasks.

From a remote sensing perspective, informal settlements' morphological characteristics, such as tiny roofs, fewer floor counts, high densities, limited space, narrow roads, and little vegetation, are studied to identify UIS. The advent of very-high-resolution (VHR) satellites provides remote sensing images with a spatial resolution of 1 m per pixel or better, providing a crucial basis for discriminating tiny, heterogeneous geological objects, such as buildings and roofs. Thus, in the context of UIS mapping, increased attention to utilizing VHR satellite images can be observed in recent years, together with the advent of more novel and sophisticated image analysis techniques, such as object-based image analysis (OBIA) [17], [18], [19], ML-based methods [2], [20], [21], or, more recently, deep learning-based (DL-based) methods [1], [22], [23].

Traditional pixel-based image classification assigns a category label per pixel. All pixels are the same shape, same size, and have no concept of their neighbors. OBIA-based methods [24], [25] segment an image by grouping small pixels into objects instead of on a per-pixel basis. OBIA-based methods classify objects in remote sensing images based on geometry, color, area, shape, texture, spectral information, etc., and they achieve remarkable results in parcel-scale or block-scale UIS mapping tasks. However, OBIA-based methods rely on fine-tuning segmentation parameters and a classifier to assign the segmentation results to the target classes. It is difficult and time-consuming to determine proper fine-tuning segmentation parameters to satisfy various objects with different scales. This results in unsatisfactory performance when using OBIA-based methods for large-area and fine-scaled UIS mapping.

Large-scale UIS mapping is mainly based on ML-based methods [26], [27], [28], which aim at generalizing specific semantic information in remote sensing images from labeled categories to provide transferability of the learned classifier to recognize UIS areas from unlabeled samples. Most ML-based methods work well when their human-designed features are adequate and representative. In this case, ML-based methods are good at optimizing parameters to update the classifier to make optimal predictions. Studies in UIS mapping indicate that contextual image features, such as morphological profiles [29], [30], physical appearances, and the gray-level cooccurrence matrix (GLCM) [31], [32], are key features, and they are used extensively in UIS mapping in various ML-based models, such as random forests (RFs) [33], [34], support vector machines (SVMs) [35], and decision trees (DTs) [28]. However, manually designed features are often incomplete, over-specified, and costly to obtain and validate. In addition, traditional ML-based models sometimes suffer poor accuracy when the surrounding environments are complex due to the inability of these models to fit complex nonlinear equations.

DL-based methods attempt to train a robust nonlinear deep neural network to automatically learn multiple levels of feature representations exclusively from its input data without the requirement for additional user input. Recent advancements in the field have demonstrated that DL has superior feature learning and transfer capability for solving highly computational tasks, and it even outperforms human interpretation in computer vision and natural language processing tasks. Among various DL-based models, studies have noted that variances of convolutional neural networks (CNNs) [36], [37], [38], such as VGGNet [39], [40] are extensively exploited in parcel-based or block-based UIS mapping tasks. These research studies consider UIS mapping an image classification problem and feed models with a parcel or block of remote sensing images, and the output is a category that enables determining whether informal settlements are contained.

Image classification in a parcel or block scale cannot meet the requirement of fine-scale UIS mapping [2], [27]; Therefore, semantic segmentation methods are studied to understand UIS at the pixel level. Semantic segmentation refers to the task of assigning a class label to every pixel in the image, i.e., each pixel of an image is assigned a semantic category, and this differs from the parcel- or block-based image classification methods that generally assign a semantic category to groups of pixels within the block and parcel. Therefore, the semantic segmentation methods are more refined than parcel- or block-based image classification methods and can take into account the differences in the objects within the parcels. Among them, variances of fully convolutional networks (FCNs) [41], [42], such as Unet [43], [44], DeepLab series [45], [46], and other semantic segmentation models [47], [48], have achieved better fine-scaled results in UIS mapping, both in accuracy and time-cost, than previous image classification methods. However, these DL-based methods still have room for improvement.

In some UIS mapping studies [17], [18], [19], a connected region with similar or homogeneous characteristics was defined as a basic unit. Previous studies have adopted parcels or blocks, relatively homogeneous regions that differ from their surrounding environments, as the basic units for image classification of UIS [2], [27]. Some studies treat UIS mapping as a semantic segmentation task and obtain the final UIS mapping result by classifying each pixel in the area [22], [23]. However, few studies utilize object-level information to improve UIS mapping. Object-level information has been studied to assist parcel- and pixel-level information in various remote sensing mapping tasks [49], [50]. In UIS mapping, informal settlements are formed by buildings (objects). Thus utilizing object-level information is possible to improve UIS semantic segmentation. Furthermore, current UIS mapping studies mainly use single-modality remote sensing images to identify UIS areas. Using only a single modality of remote sensing image features limits the effectiveness and efficiency of informal settlements' semantic segmentation. Multimodal data, such as geospatial data, can provide rich information that cannot be obtained directly from remote sensing images (such as building height, building area, etc.); thus, they can be used in auxiliary UIS semantic segmentation tasks and obtain more accurate and fine-scaled semantic segmentation results.

Aiming at fine-scale UIS mapping results, this article proposes a UIS semantic segmentation method, namely UisNet, that utilizes a transformer-based block to separately model the multimodal data, including high-spatial-resolution remote sensing (HRRS) images (parcel- and pixel-level information)

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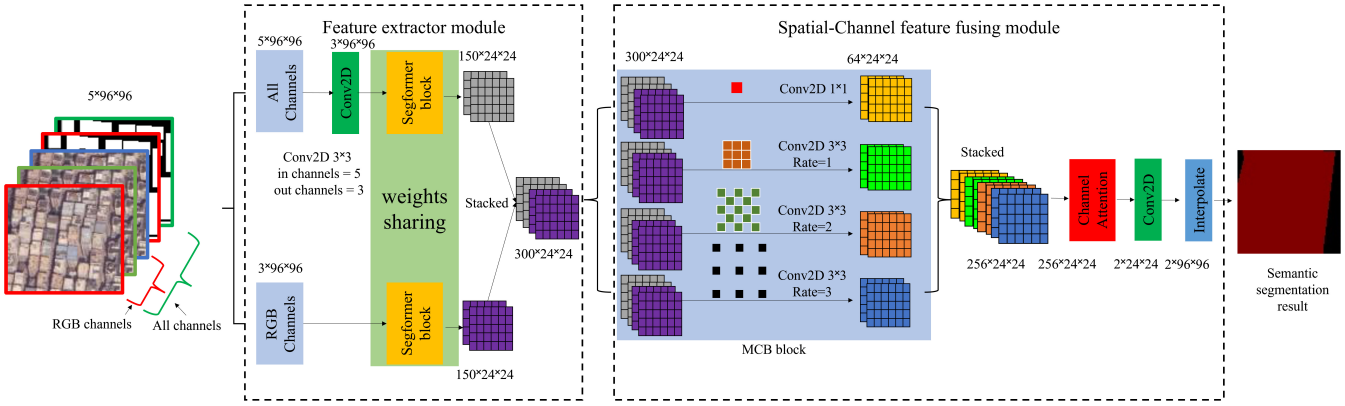

Fig. 1. Architecture of the proposed UisNet.

and rasterized building polygon data (object-level information). Then, a spatial-channel feature fusing module is added to fuse the output features from the multimodal data to improve the semantic segmentation results. The proposed method differs from the segmentation transformers presented in the computer vision (CV) context in two aspects. One is that the proposed method focuses on multimodality semantic segmentation by using two weight-sharing Segformers to extract features from different modalities. In contrast, existing segmentation transformers mainly employ single modality data. The other one is that traditional transformers do not contain convolution-based layers except the patch-embedding layer. In this article, we use the atrous convolution-based multiscale context block (MCB) and the channel attention module to extract spatial and channel features in the proposed method to get better fusion results.

The experiments were conducted in Shenzhen City and a public and widely used gaofen image dataset (GID) dataset. The proposed UisNet achieved superior performance compared to other state-of-the-art semantic segmentation models. The main contributions of this article are summarized as follows.

1) Aiming at UIS semantic segmentation tasks, a transformer-based, multimodal fusion neural network, namely UisNet, is proposed. By using the attention mechanism and the patch division and fusion process of the transformer block, the local and global information of the input data are fully considered, and the UIS mapping result is improved.
2) By comprehensively using multimodal data, including remote sensing images and building polygon data, the accuracy of UIS semantic segmentation is improved.
3) By using the proposed spatial-channel feature fusing module to capture spatial and channel features simultaneously and fusing the parcel and pixel information of remote sensing images and the object information from the building polygon data, the accuracy of UIS semantic segmentation is improved.
4) We assess the UIS semantic segmentation uncertainties by analyzing the mIoU and the UIS area proportions at the parcel scale, and find that as the proportion of UIS areas increases, the performance of UIS semantic segmentation improves, and the accuracy uncertainty of the proposed model decreases.

The rest of this article is organized as follows: Section II outlines the proposed method, Section III describes the experimental results, and Section IV presents the conclusions.

II. Proposed Method

This article introduces UisNet, a UIS semantic segmentation method that utilizes a transformer-based block to receive multimodal data, including HRRS images and building polygon data, to identify UIS. Formally, an HRRS image, together with building polygon data $I = \{x_{ij}\}$, was given, where i denotes the horizontal pixel index of the image I and j denotes the vertical pixel index of the image I. The output is $Y = \{y_{ij}\}$ where y_{ij} denotes the UIS label of x_{ij}.

The multimodality of the input data are raster HRRS images (Tif format) and building polygon data (Shapefile format). The building height information, area information, and building outlines which can be obtained from the building polygon data, are discriminative for distinguishing UIS from other areas. Specifically, UIS typically has low building heights, limited areas, and crowded buildings. Moreover, for the UIS semantic segmentation task, it is challenging to extract the geometries of UIS buildings accurately. Therefore, adding object-level building outlines from building polygon data can help improve the accuracy of UIS semantic segmentation. The above reasons are the motivation for selecting building polygon data for UIS semantic segmentation. In the building polygon dataset, there are building outlines and the floor information which include floor numbers and building area, were both used in this study. The building outlines are first utilized to obtain the building mask, and then the value of floor numbers and building area are employed to assign values to the pixels of the building mask. By the help of this, the building mask can generate input data with the building height channel and the building area channel.

The proposed UisNet, shown in Fig. 1, includes a feature extractor module and a spatial-channel feature fusing module. The pseudocode of the UIS semantic segmentation method in a PyTorch-like style is shown in Algorithm 1. In the same way, the feature extractor module uses two same weight-sharing

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Algorithm 1 Pseudocode of the Segmentation Method in a PyTorch-Like Style

Input: Remote sensing images: $I = \{i_1, i_2, \ldots, i_N\}$, Rastered building data: $B = \{b_1, b_2, \ldots, b_N\}$. Label: $L = \{l_1, l_2, \ldots, L_N\}$

```
for i, b, l in I, B, L: do
    ALL_channels = torch.cat([i, b], dim = 1)
    RGB_channels_features = Segformer(i)
    ALL_channels_features = Segformer(Conv2D(ALL_channels))
    fuse = torch.cat([RGB_channels_features, ALL_channels_features], dim = 1)
    fuse = MCB(fuse)
    fuse = Channel Attention(fuse)
    fuse = Conv2D(fuse)
    fuse = Interpolate(fuse)
    loss = OhemCrossEntropy(fuse, l)
    loss.backward()
    parameters update
end for
```

SegFormer [51] blocks to extract the different features of the red-green-blue (RGB) channels (RGB channels of HRRS images) and all input channels (RGB channels of remote sensing images together with building height channel and the building area channel) and obtain the learned feature maps. The reason for putting the RGB channels and all input data channels into two weights-sharing Segformer models is that this architecture can balance the characteristics of multimodal data and obtain results with good qualities. In this study, remote sensing images (the RGB channels) are important for the UIS semantic segmentation. Meanwhile, the building polygon channels (the building height channel and the building area channel) are also important for capturing object-level information, which is difficult to be captured from the RGB channels directly. Therefore, we stacked the RGB channels and the building polygon channels into all input data channels and put the RGB channels and all input data channels into two weights-sharing Segformer models. The weights-sharing strategy is used for reducing the parameters of UisNet.

The spatial-channel feature fusing module uses an atrous MCB module that utilizes atrous convolution with different rates to enhance the spatial features and receive feature maps with multiscale spatial information and a channel attention module, which uses the channel attention mechanism to compute channel-wise attention on the feature maps from the multiscale feature learning module to learn the feature weights between different channels at the same spatial location and then produce the final UIS semantic segmentation maps. The detailed architecture of the proposed UisNet is introduced in the following two subsections.

A. Feature Extractor Module

High-resolution spatial features, such as detailed information about land surfaces, ground objects, and buildings, have been proven to enhance UIS mapping. Numerous studies have shown that CNNs can efficiently learn spatial features for the shared convolution kernel and shifting invariant characteristics. For example, ResNet [52] is good at learning high-level and deep spatial representations, with the benefits of a skip-connection and shortcut design. Thus, the ResNet series

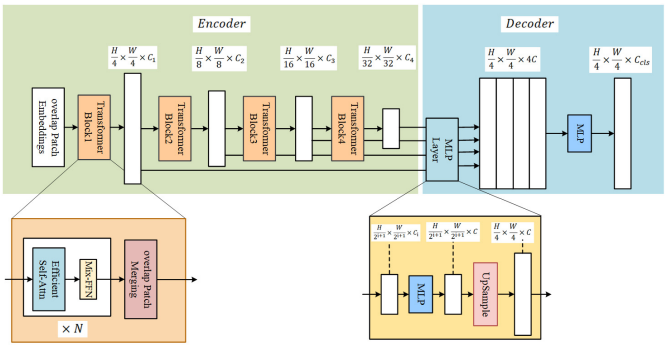

Fig. 2. Architecture of Segformer [51].

(e.g., ResNet18, ResNet50, ResNet101) is widely used as feature extractors in many semantic segmentation tasks. However, the receptive field of convolution operation is limited, meaning it can only capture the feature within a small spatial region of an HRRS image. To preserve the global context information of the input image, hundreds of convolution layers must be stacked. However, in the deep layers of CNNs, the detailed information of the input image is lost, weakening the representativeness of the extracted features and creating numerous model parameters and extensive time-costs for training.

Researchers have introduced the transformer model [53], [54] in CV tasks to solve the shortcomings of CNNs using a self-attention (SA) mechanism [55]. Transformers have a robust local-global feature learning capability due to combining patch embeddings and multilayer perceptron (MLP) layers. Patch embeddings in the first layer can learn the local information of the input image, while the MLP layers can model the global information. The transformer model can also simultaneously recognize multiple key-point information data by using a multihead attention structure. Therefore, transformer-based models are appropriate for promising performance in capturing valuable information from HRRS images and are studied with great success in various remote sensing tasks [56], [57], [58].

Thus, we adopted a widely used transformer-based model, namely the Segformer [51], as a feature extractor to capture discriminative local and global spatial features. The Segformer structure is illustrated in Fig. 2, and it contains a series of mix

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transformer encoders (Encoder) and a lightweight all-MLP decoder (Decoder). The Encoder is used to extract large-scale, coarse-grained features and small-scale, fine-grained features. The Decoder fuses the features that are produced by the different layers in the Encoder and generates the semantic segmentation mask.

In Segformer, the Encoder takes an image of shape (C, H, W) and outputs a feature map of shape $(C_4, H/32, W/32)$ using four stacked transformer blocks. Each transformer block contains an Efficient Self-Attn, a Mix-feedforward network (FFN), and an Overlapped Patch Merging module. The main computation bottleneck of the transformer block is the SA layer. In the original multihead SA process, each of the heads Q, K, and V have the same dimensions of $N \times C$, where N is the length of the sequence, $N = H \times W$. SA is defined as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{\text{head}}}}\right)V. \tag{1}$$

In the Efficient Self-Attn layer, a reduction ratio R is used to reduce the length of the sequence as follows:

$$\hat{K} = \text{Reshape}\left(\frac{N}{R}, C \cdot R\right)(K) \tag{2}$$

$$K = \text{Linear}(C \cdot R, C)\left(\hat{K}\right) \tag{3}$$

where $\text{Reshape}((N/R), C, R)(K)$ refers to reshape K to one tensor with a shape of $(N/R) \times (C \cdot R)$. $\text{Linear}(i, j)$ refers to a linear layer taking an i-dimensional tensor as input and outputting a j-dimensional tensor.

The output of the Efficient Self-Attn layer is then fed to the Mix-FFN layer. The Mix-FFN layer considers the effect of zero padding to leak location information by directly using a 3×3 convolutional layer ($\text{Conv}_{3\times3}$). The Mix-FFN layer is defined as

$$x_{\text{out}} = \text{MLP}(\text{GELU}(\text{Conv}_{3\times3}(\text{MLP}(x_{\text{in}})))) + x_{\text{in}} \tag{4}$$

where x_{in} is the feature from the Efficient Self-Attn layer. x_{out} is then fed into the Overlapped Patch Merging operator to obtain hierarchical feature maps. The feature outputs of the transformer blocks of different depths together form the Encoder output x^i_{out}.

The Encoder output is then fed into the Decoder, which uses a lightweight All-MLP layer to enlarge the feature map to the shape of $(C_{\text{cls}}, H/4, W/4)$. The All-MLP Decoder consists of four main steps that are defined as follows:

$$x^i_{\text{out}} = \text{Linear}(C_i, C)\left(x^i_{\text{out}}\right) \tag{5}$$

$$x^i_{\text{out}} = \text{Upsample}\left(\frac{W}{4} \times \frac{W}{4}\right)\left(x^i_{\text{out}}\right) \tag{6}$$

$$x_{\text{out}} = \text{Linear}(4C, C)\left(\text{Concat}\left(x^i_{\text{out}}\right)\right) \tag{7}$$

$$x_{\text{out}} = \text{Linear}(C, N_{\text{cls}})(x_{\text{out}}) \tag{8}$$

where x^i_{out} is the ith feature map of hierarchical feature maps x_{out}. N_{cls} is the number of categories in the original Segformer article [51]. However, in this article, N_{cls} is set to 150 to obtain high-dimensional features from the Segformer.

In this study, we used two same weight-sharing Segformer models (version Segformer-B2) to receive the RGB channels and all input data channels, respectively. Specifically, for the RGB channels (shape $(3 \times 96 \times 96)$), we fed the input data into a Segformer to get a feature map of shape $(150 \times 24 \times 24)$. For all channels (shape $(5 \times 96 \times 96)$), we first fed the input data into a two-dimension CNN with a 1×1 convolution and 3 convolution kernel to reduce their dimension to 3 and obtain a feature map of shape $(3 \times 96 \times 96)$. Then, the acquired feature map was fed into the Segformer to get a feature map output of shape $(150 \times 24 \times 24)$. The above two feature maps of shape $(150 \times 24 \times 24)$ were finally stacked into a feature map of shape $(300 \times 24 \times 24)$.

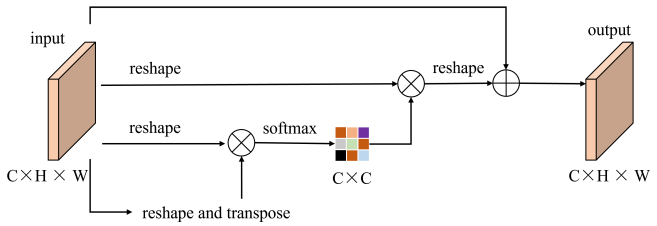

Fig. 3. Architecture of channel attention module [61].

B. Spatial-Channel Feature Fusing Module

The proposed spatial-channel feature fusing module is shown in Fig. 1. In the spatial-channel feature fusing module, an MCB module [59] with atrous convolution kernel [60] is first adopted to increase the receptive field of the convolution kernel to effectively capture multiscale contextual information.

Through the feature extractor module, a stacked feature map with a shape of $(300 \times 24 \times 24)$ was generated, and then the MCB was attached after the feature map. In the MCB, the atrous convolution kernel allows different size expansion factors to receive different receptive field sizes to capture multiscale contextual features. In the MCB, three copies of the feature maps from the feature extractor module were separately fed into three branches of atrous convolutions with 64 convolution kernels of size 3×3 and different dilated rates $(1, 2, 3)$ separately. Such that variant scales of the feature maps can be extracted. One copy of the feature maps was passed into a 1×1 convolution layer with 64 convolution kernels. Then, the four output feature maps were stacked to generate a feature map of a shape $(256 \times 24 \times 24)$.

After the MCB block, each channel map of high-level feature maps from the MCB block could be viewed as a class-specific response associated with each other. We then used the channel attention module [61] to improve the class-specific feature representation by exploiting the interdependencies between the obtained channel maps. The structure of the channel attention module is explained in Fig. 3. The stacked feature map of shape $(256 \times 24 \times 24)$ produced by the MCB block was fed into a channel attention module [61] to obtain an enhanced class-specific feature map, with the shape of $(256 \times 24 \times 24)$. After the channel attention module, a two-dimension CNN with a 1×1 convolution and 2 convolution kernels was added to transform the input to an output of shape $(C \times 24 \times 24)$, where C means the number of categories. C was set to 2 in this article. Finally, we used bilinear interpolation

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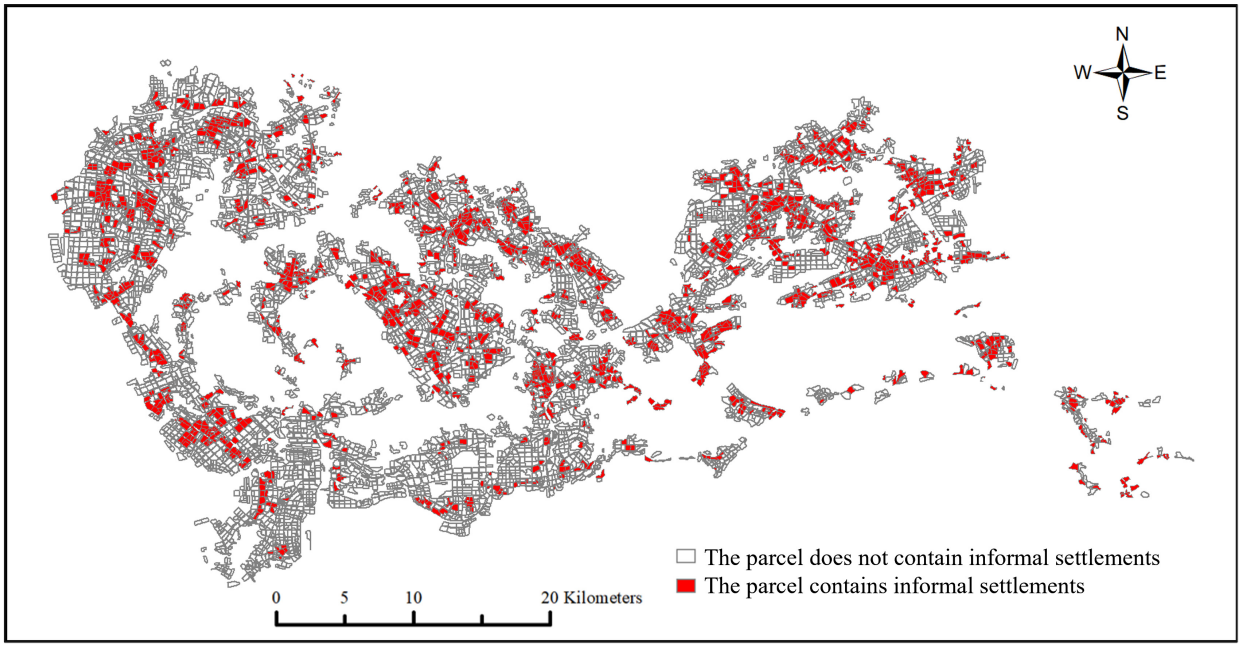

Fig. 4. Parcels of Shenzhen City.

to enlarge the output to the same size as the input image, that is $(C \times 96 \times 96)$.

III. Experiment and Analysis

To validate the effectiveness of the proposed UisNet, we designed several experiments in Shenzhen City and the GID dataset that used the proposed UisNet, as well as other state-of-the-art semantic segmentation methods. Section III-A introduces the study area. Section III-B presents an overview of the information and preprocessing stage of the multimodalities data. Section III-C contains the results and analysis of the semantic segmentation of UIS. Section III-D contains the analysis of UIS area proportion in Shenzhen City. To demonstrate the generalization of the proposed method, we conduct experiments on a widely used public dataset, the GID dataset. The experiments and comparison on the GID dataset is presented in Section III-E.

A. Study Area

We conducted UIS semantic segmentation in Shenzhen City, which is China's first special economic zone located in the southern part of Guangdong Province. Over the past three decades, Shenzhen City has been one of the fastest-growing cities in the world, and it covers an area of approximately 2,020.5 km^2. Many old, dilapidated, and tiny informal settlements still exist in Shenzhen City and must be improved. Therefore, we selected Shenzhen City as a study area for UIS semantic segmentation.

B. Experimental Data Preprocessing

Many studies have indicated that multimodal remote sensing and geospatial data are beneficial in distinguishing highly heterogeneous urban scenes [62], [63], [64], [65]. In the real-world urbanization process, land use parcels serve as the basic unit of urban settlements' formation and development. Therefore, we first investigated urban settlements from the perspective of land use parcels. As shown in Fig. 4, we divided Shenzhen City into 6618 land use parcels based on the road network.

For each parcel, we first judged whether it contained informal settlements. Then, if the parcel had informal settlements, pixel-level labeling of the informal areas was conducted for UIS semantic segmentation. In this way, we obtained 1618 parcels that contained informal settlements (as shown in the red parcels in Fig. 5). Then, we obtained the semantic segmentation dataset of informal settlements through pixel-level labeling on the LabelMe (`https://github.com/wkentaro/labelme`) platform. Finally, we obtained a semantic segmentation dataset of informal settlement areas in Shenzhen City (UIS-Shenzhen dataset), including a total of 1618 labeled samples. Some of the UIS samples are shown in Fig. 6.

TABLE I
Multimodalities Data Used

Data Type	Time	Spatial Resolution	Data source
HRRS images	2020	$1.19m$	Google Earth
Building polygon data	2020	-	Baidu Map

The multimodal remote sensing and geospatial data are represented in Table I. These two kinds of data are registered in the data preprocess stage. The coordinate system is unified at first and then the offset registration is performed for the two modalities of data. The following subsections further describe

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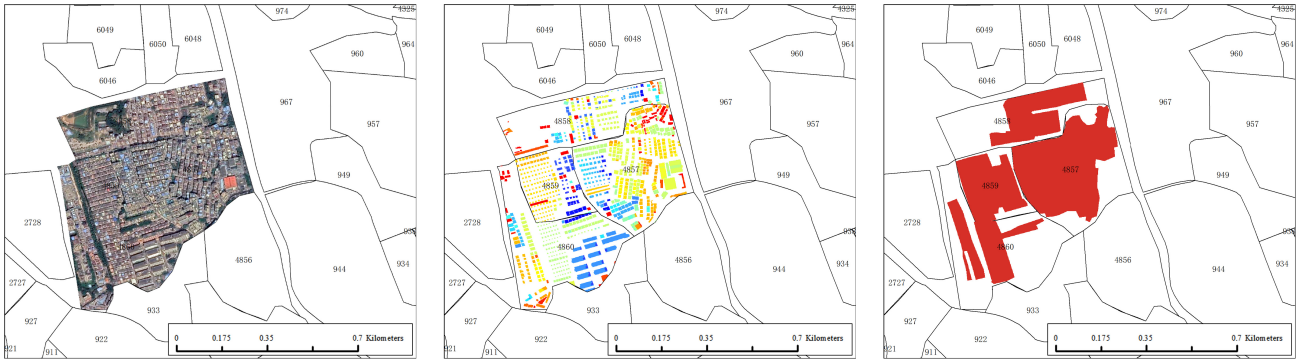

Fig. 5. Illustration of multimodalities remote sensing and geospatial data with the corresponding label. (a) HRRS images. (b) Building polygon data. (c) Informal settlements label.

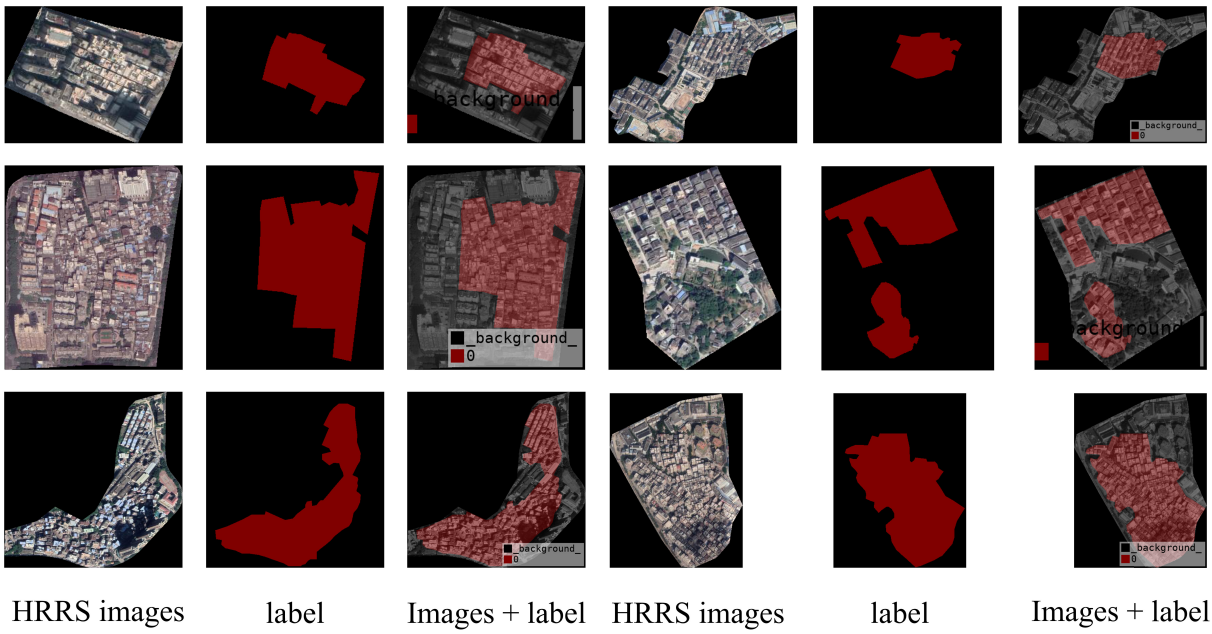

Fig. 6. Some samples of UIS in Shenzhen City.

the preprocessing steps that we applied to the multimodal remote sensing and geospatial data.

1) HRRS Images Preprocessing: As presented in Fig. 5(a), we retrieved HRRS images from Google Earth in 2020 with a spatial resolution of 1.19 m. We cropped these HRRS images according to their land use parcel boundaries. We collected a total of 6618 HRRS images, in which 1618 of these parcels contained informal settlements.

2) Building Polygon Data Preprocessing: As shown in Fig. 5(b), we obtained building polygon data from Baidu Maps. The original building polygon data contained 670 464 polygon planes of buildings in Shenzhen City. Each polygon had corresponding floor information, which indicated the number of floors in the building. To satisfy the input requirements of the model, we rasterized the polygon floor data using the value of floor numbers and area information. The resolution was consistent with the HRRS images, which was 1.19 m.

C. Analysis and Results of UIS Semantic Segmentation

During training, we used the Adam optimization method [66] and set the learning rate to 0.0001. We set the batch size to 64 and trained a total of 100 epochs. All implementations were based on PyTorch (Version torch == 1.8.2), TorchVision (Version torchvision == 0.9.2), and Python (Version Python == 3.8.12). We employed an NVIDIA RTX 3090 graphics processing unit (GPU). The loss function was OhemCrossEntropy [67]. We evaluated the results of the compared models using mean intersection over union (mIoU) and overall accuracy (OA), which are broadly accepted as evaluation criteria in semantic segmentation tasks.

To facilitate comparisons with other models, we randomly split the total 1618 samples as follows: 80% as the training set (1,294), 10% as the validation set (162), and 10% as the testing set (162). The spatial distribution of the training, testing, and validation set is shown in Fig. 7. Due to the diverse

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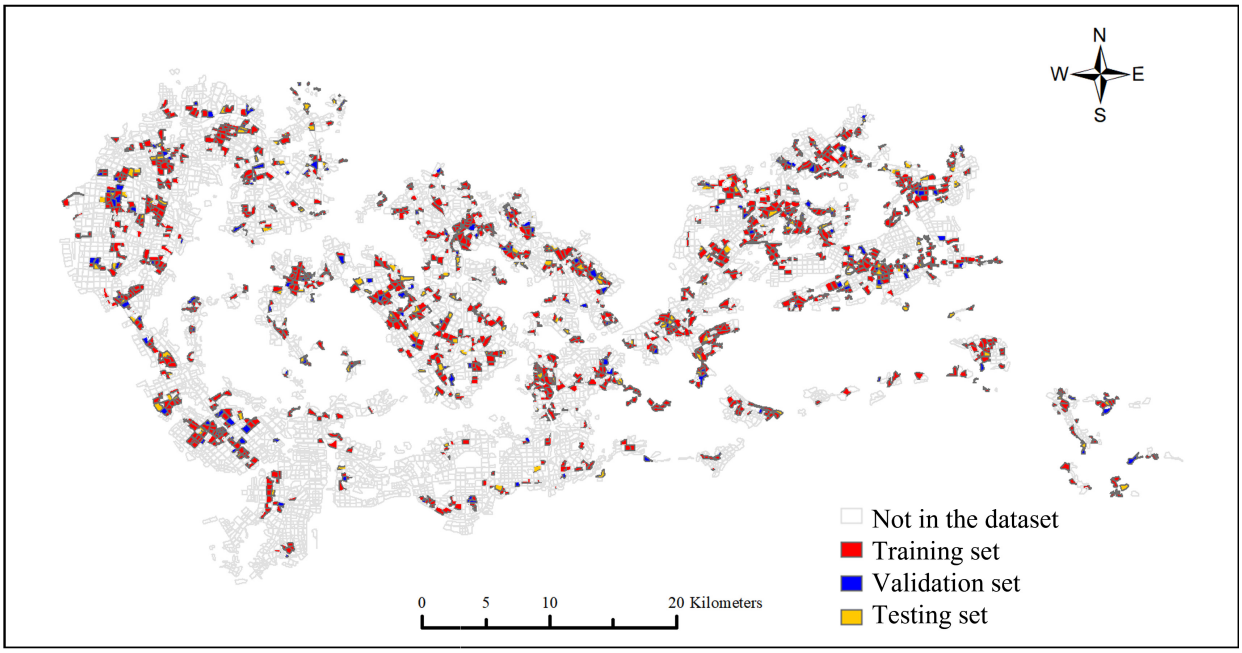

Fig. 7. Spatial distribution of the training, testing, and validation set.

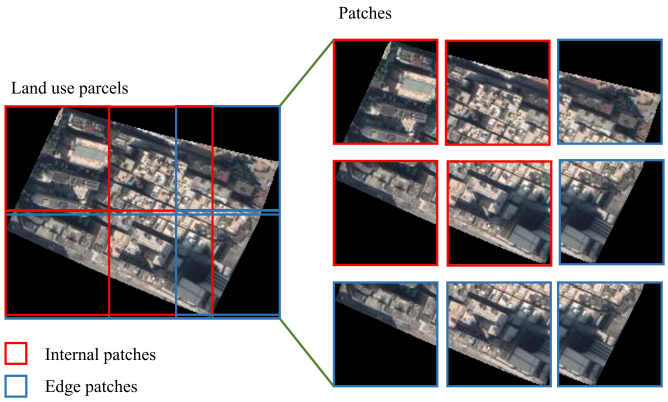

Fig. 8. Illustration of the sliding window.

size of the parcels, it is unlikely for a fixed model to receive samples of different sizes. To feed these samples to a model, for each parcel in the training set, validation set, and testing set, we applied a sliding window to split the whole parcel into patches of 96 pixels in length and 96 pixels in width. An illustration of the sliding window is presented in Fig. 8. In this way, we divided the training set into 27 952 patches, the validation set into 3315 patches, and the testing set into 3389 patches.

To validate the effectiveness of the proposed UisNet, we designed multiple comparative experiments using several semantic segmentation models, including FCN, UNet, DeepLab series, and Segformer. In Table II, all comparison models, including UNet, FCN, Deeplab-V3, and Segformer, use only remote sensing images. Table II shows the mIoU and OA of different models for UIS semantic segmentation on the validation set and the testing set. The results have

TABLE II
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED UISNET

		Validation Set		Testing Set		Parameters Amount
		OA (%)	mIoU	OA (%)	mIoU	
Comparison models	UNet	94.08	83.92	93.92	83.52	17.3M
	FCN	94.44	84.63	94.42	84.54	32.9M
	Deeplab-V3	94.47	84.82	94.38	84.55	39.6M
	Segformer	94.53	84.91	94.53	84.84	27.5M
The proposed method	UisNet-1	94.69	85.37	94.68	85.27	28.0M
	UisNet	**94.89**	**85.78**	**94.80**	**85.51**	28.0M

two parts. First, we conducted several experiments on the comparison models. Second, we tested the UisNet model. Then, we compared all results with the proposed UisNet. All the versions of the UisNet used pretrained weights on the Cityscapes dataset [68] as their initialization weights.

The OA and mIoU of the compared models and the proposed UisNet are presented in Table II. Of the compared models, the DeepLab-V3 (version deeplabv3_resnet50 model in the torchvision.models.segmentation library) and FCN (version fcn_resnet50 model in torchvision.models.segmentation library) had the highest OA and mIoU; however, the number of parameters of the DeepLab-V3 model was more than that of the FCN model. The UNet model had the lowest OA and mIoU and the fewest parameters. Segformer had a similar OA and mIoU to DeepLab-V3 and FCN. However, the number of parameters for SegFormer was much less than that of DeepLab-V3 and FCN. This indicates that using SegFormer as the backbone in the proposed method can efficiently utilize the model parameters and obtain similar results with fewer parameters. The proposed UisNet had the highest OA and mIoU compared to all the compared models in both the validation set and the testing set, with an improvement of 0.5% and about 1%

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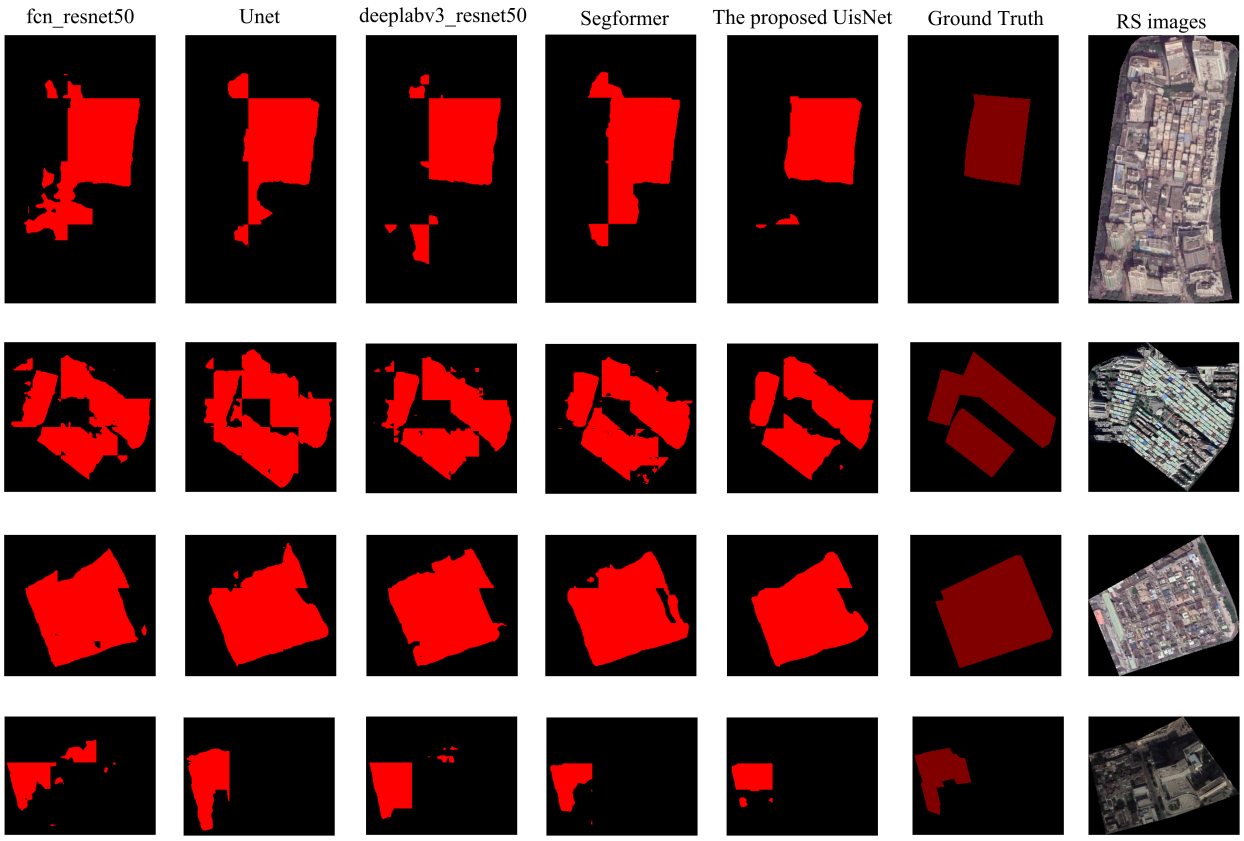

Fig. 9. Illustrations of some UIS semantic segmentation results.

TABLE III
ABLATION STUDY OF THE PROPOSED METHOD

Method	Segformer	MCB	Channel Attention	OA (%)	mIoU
Segformer (RS)	√			94.53	84.84
UisNet (RS)	√	√	√	94.56	85.01
Segformer	√			94.68	85.27
UisNet-a	√	√		94.68	85.17
UisNet-b	√		√	94.72	85.43
UisNet-c	√	√	√	**94.80**	**85.51**

in OA and mIoU, respectively. Evidently, fusing multimodality data (HRRS images and building polygon data) greatly enhanced the UIS semantic segmentation results. UisNet-1 is a comparative model that does not contain the spatial-channel feature fusing module in UisNet. By adding the spatial-channel feature fusing module, UisNet can further improve the mIoU and OA by 0.24 and 0.41 compared to UisNet-1, respectively, while the increased number of parameters can be ignored.

We provide some UIS semantic segmentation results of the comparison models in the testing set in Fig. 9. The proposed UisNet can more accurately segment informal residential areas. Compared with other methods, our results have fewer misclassified pixels and better maintain the geometric features of the objects in the segmentation results (flatter edges, more regular shapes, etc.). For example, the FCN results contain many internal holes, while the Unet and DeepLab-V3 results include many misclassified salt and pepper noises outside of the informal settlement areas. The SegFormer model easily misclassified the background parts into informal settlements. Only the proposed UisNet can accurately segment informal settlements, and this demonstrates the superiority of the proposed UisNet in UIS semantic segmentation.

An ablation study is used to evaluate the effectiveness of various components in the proposed UisNet. The experimental results (in the testing dataset) are shown in Table III. Segformer (RS) and UisNet (RS) represent models that only employ remote sensing images as input. Segformer, UisNet-a, UisNet-b, UisNet-c all take multimodality data as input. Segformer (RS), which achieved the best performance among the comparison models in Table II. It is utilized as the baseline model in this ablation study. Other models contain at least one

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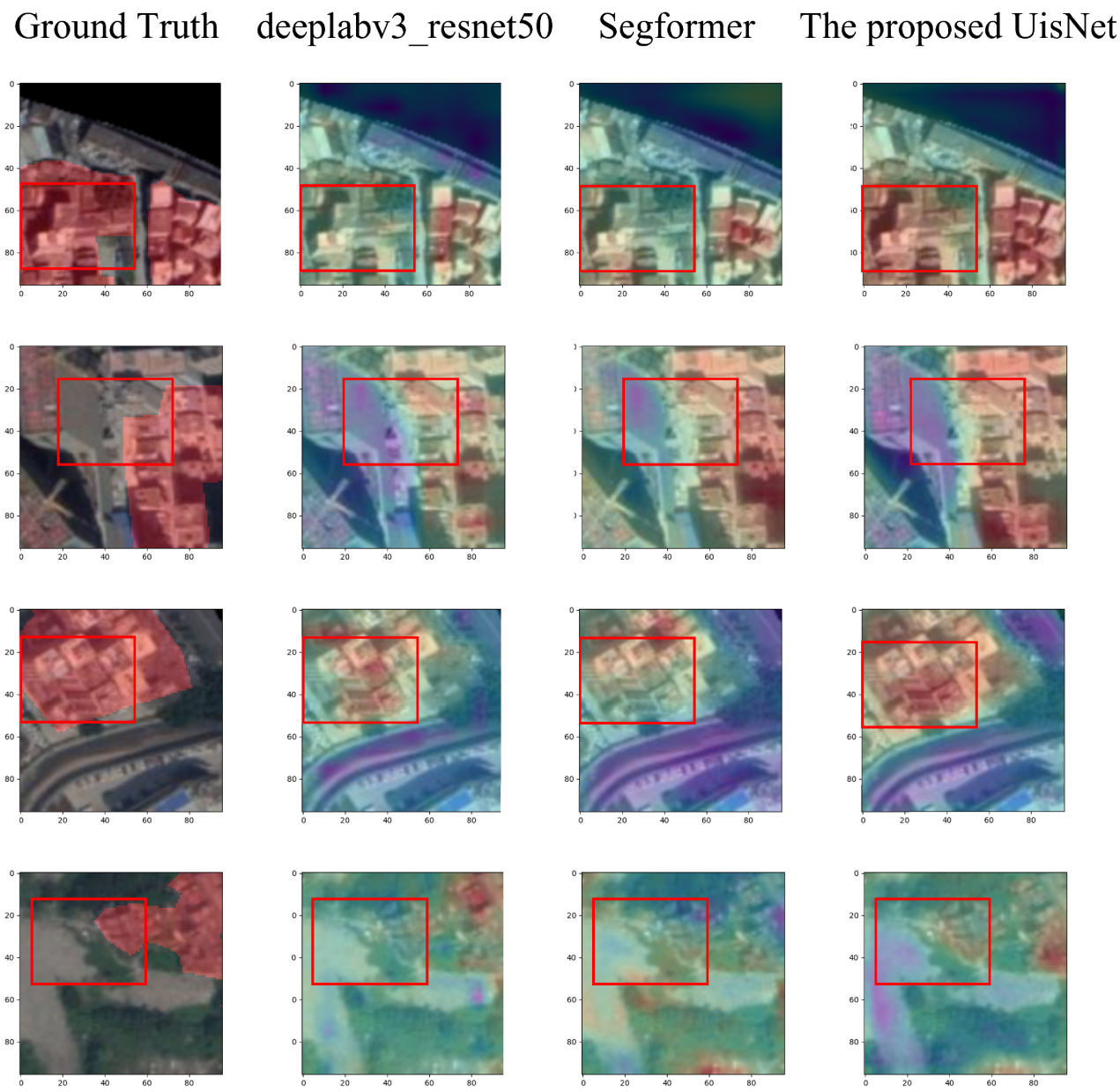

Fig. 10. Class-specific attention maps over the images.

key components which can be Segformer, MCB, or Channel Attention.

It can be observed that the Segformer (RS) with only remote sensing images achieved the least significant OA of 94.53% and a mIoU of 84.84%, which is less than the Segformer that takes multimodality data as input, with a decrease in OA of 0.15% and a mIoU of 0.37%. This indicates the superior of fusing remote sensing and building polygon data in UIS classification. The UisNet (RS) with only remote sensing images achieved an OA of 94.56% and a mIoU of 85.01%, which outperformed the Segformer (RS). This shows the superiority of the proposed UisNet with the MCB module and the channel attention mechanism compared to the Segformer (RS) using only remote sensing images. UisNet-a and Segformer achieved similar OA and mIoU, demonstrating that only adding the MCB model cannot significantly improve the performance. UisNet-b outperformed Segformer with mIoU of about 0.2%, showing that adding the Channel Attention block enhanced the performance of the model. By combining the MCB and the Channel Attention block, UisNet-c achieved the best performance with an OA of 94.80% and a mIoU of 85.51%. This indicates that the network structure of using the MCB module and the channel attention mechanism is effective in fusing multimodal data and can achieve better UIS semantic segmentation performance.

In the proposed UisNet, multimodality data and the proposed network structure both played essential roles. In Table III, it can be observed that the UisNet (RS) with

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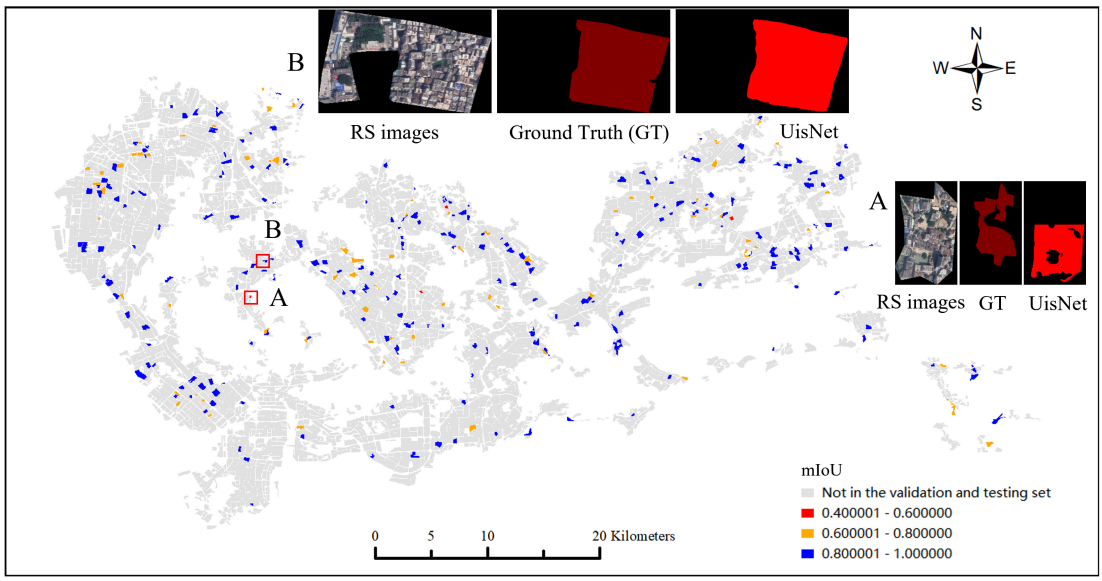

Fig. 11. mIoU of parcels in Shenzhen City in the testing and validation set. "A" denotes an example parcel with low mIoU, and "B" denotes an example parcel with high mIoU.

only remote sensing images achieved an OA of 94.56% and a mIoU of 85.01%, which is worse than the UisNet-c that takes multimodality data as input, with a decrease in OA of 0.24% and a mIoU of 0.50%. It shows that multimodality data can help improve a mIoU of 0.50% compared to the model using only remote sensing images. UisNet-c outperformed Segformer with a mIoU of 0.24%. It indicates the proposed network structure can help to improve a mIoU of 0.24% compared to the baseline model without the MCB and the Channel Attention block. In general, in the proposed UisNet, multimodality data and the proposed network structure both played essential roles.

Note that the UisNet (RS) using only remote sensing images outperformed the Segformer (RS) with a mIoU improvement of 0.17% in Table III, while the UisNet-c outperformed the Segformer using multimodality data with a mIoU improvement of 0.24%. This indicates that the network structure of using the MCB module and the channel attention mechanism is more effective in fusing multimodal data and can achieve better UIS classification performance by simultaneously modeling spatial and channel features from multimodal data.

Furthermore, as shown in Fig. 10, we used visual class-specific attention maps to display the position of class-specific regions located by semantic segmentation models. The attention maps were obtained from the feature map before the softmax classifier layer. The attention maps contain two channels, which correspond to Others and UIS, respectively. We display the channel that corresponds to the UIS categories as the class-specific attention map. In the class-specific attention map (shown in Fig. 10), the red color indicates the high confidence in UIS, and the purple color indicates the opposite.

It can be observed that the class-specific attention maps in our method is the most similar one to the ground truth, which is reflected by the more accurate recognition of the UIS areas. In the feature map, the areas with high confidence (shown in red) of the UisNet model are quite close to the ground truth. In addition, in the class-specific attention maps of our method, the class-specific attention maps in different regions are more distinct in purple and red by comparing to other models. This indicates that our model has the better discriminative ability for UIS (Red) and Others region (Purple). In contrast, there are large number of regions between red and purple (such as yellow and blue) in the class-specific attention maps of DeepLab-V3 and Segformer. It indicates that these models have less confidence in identifying these regions which make them have difficulties to identify the categories accurately. It can be observed from the above class-specific attention maps visualization results that the proposed UisNet can more accurately identify the categories, geometric shapes, and positions of UIS areas and has superior performance to other models.

The results of the UIS semantic segmentation in the validation and testing set are shown in Fig. 11. In Fig. 11, only four parcels of the 324 parcels has a mIoU value between 0.4 to 0.6. There are 76 parcels has a mIoU value between 0.6 to 0.8, and 244 parcels has a mIoU value between 0.8 to 1. From the perspective of mIoU spatial distribution, it can be observed that the parcels with a high mIoU which is greater than 0.8 are mainly localized in the western, southern coastal regions, and northeast regions of Shenzhen City. From the perspective of the geometry and visual appearance of the buildings, it can be found from A and B in Fig. 11 that the parcels with high mIoU (like B) contain buildings with clear geometric shapes while parcels with low mIoU (like A) contains buildings that block each other and have indistinguishable boundaries.

D. Analysis of UIS Area Proportion and Semantic Segmentation Uncertainty

We calculated the proportion of UIS area at a parcel scale to further characterize the UISs' spatial distribution in

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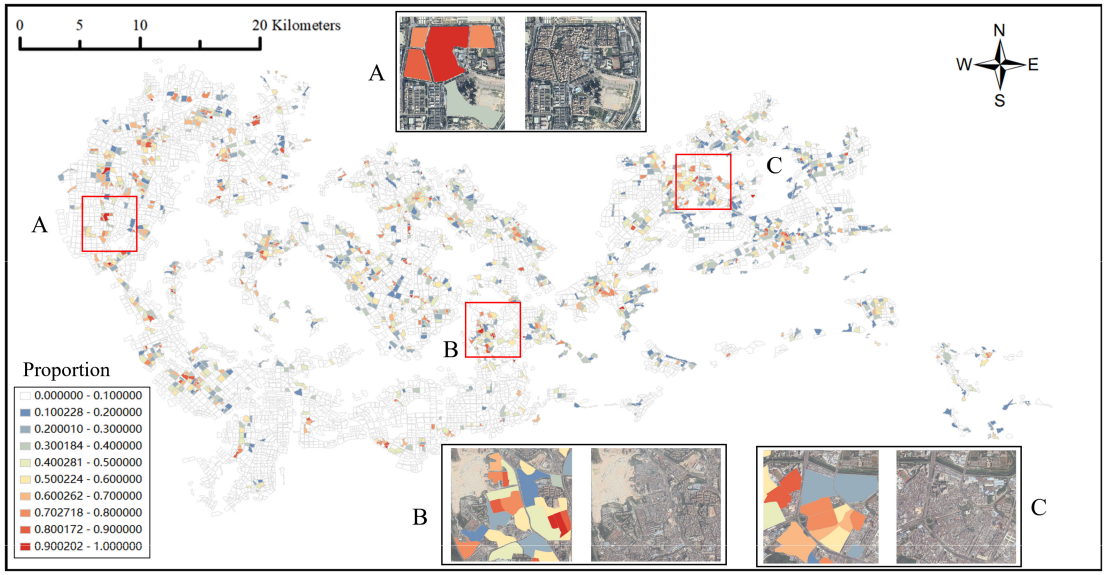

Fig. 12. Proportion of UIS area in parcels of Shenzhen City. "A,""B," and "C" denotes example regions with a high proportion of informal settlements in the northwest, central and northeast regions of Shenzhen City, respectively.

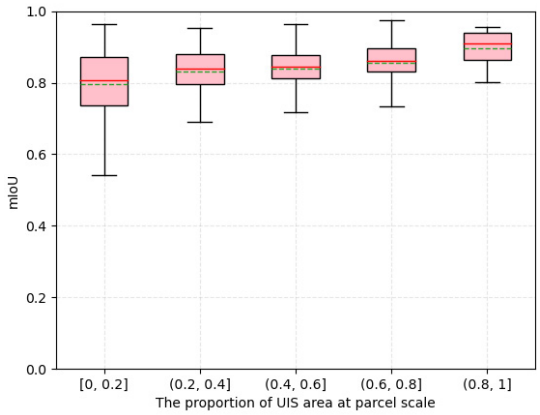

Fig. 13. Comparison of mIoU and UIS area proportion in the testing and validation set.

TABLE IV
RESULTS FOR DIFFERENT MODELS AND THE PROPOSED UISNET ON THE GID DATASET

Class	UNet	FCN	Deeplab-V3	Segformer	UisNet-RGB	UisNet
background	61.63	66.57	67.19	68.45	**68.74**	68.09
industrial land	49.27	54.57	55.54	**57.46**	56.06	55.42
urban residential	57.34	60.94	62.60	**63.78**	63.68	62.03
rural residential	46.83	50.67	52.77	52.48	**53.26**	52.79
traffic land	50.75	54.60	57.40	**58.82**	58.65	55.09
paddy field	61.89	68.66	67.97	73.44	70.33	**73.99**
irrigated land	72.36	75.96	77.72	79.28	**80.06**	79.51
dry cropland	45.03	51.33	55.99	**59.88**	53.74	53.91
garden plot	15.69	17.35	14.78	19.21	18.15	**22.95**
arbor woodland	72.69	76.30	75.00	76.28	**76.51**	75.91
shrub land	5.29	10.27	9.56	2.28	3.27	**12.77**
natural grassland	57.29	61.79	59.10	63.80	59.14	**64.42**
artificial grassland	32.07	44.36	46.40	25.97	**46.61**	42.84
river	84.32	88.2	**89.85**	88.90	88.35	88.07
lake	65.51	74.03	77.10	79.44	81.03	**82.63**
pond	64.45	67.25	70.20	68.03	67.72	**73.19**
mIoU ($\times$ 100)	52.65	57.68	58.70	58.59	59.08	**60.23**

Shenzhen City. The proportion of UIS area at the parcel scale calculates the UIS area's ratio to the entire parcel area in the 6618 parcels in Shenzhen City. We then analyzed the semantic segmentation accuracy uncertainty of the validation set and the testing set, under different UIS area proportions at the parcel scale.

1) Proportion of UIS Area at Parcel Scale: Fig. 12 shows the proportion of UIS area at the parcel scale. In most parcels of Shenzhen City, the proportion of informal settlement areas is very low, less than 10% (0.1). Only in some parcels, the proportion of informal settlement areas are greater than 80% (0.8). In terms of the spatial distribution of the proportions of informal settlement areas, the parcels with a large proportion (more than 0.6) of informal settlement areas are mainly distributed in the northwest, central, and northeast regions of Shenzhen City. The parcels with a low proportion (less than 0.1) of informal settlements are mainly localized in the western and southern coastal regions of Shenzhen City. We also highlight parcels with a large proportion of informal settlements in the northwest regions (region A in Fig. 12), central regions (region B in Fig. 12), and northeast regions (region C in Fig. 12) of Shenzhen City with remote sensing images. The images show that parcels in these regions contain relatively large dense areas, less floor heights, and relatively dilapidated buildings.

2) Analysis of Semantic Segmentation Uncertainty Under Different UIS Area Proportion at Parcel Scale: As shown in Fig. 13, UIS areas proportion denotes the ratio of the UIS area to the parcel area to which the UIS belongs. In this case, UIS area samples in the validation and testing set of Shenzhen City

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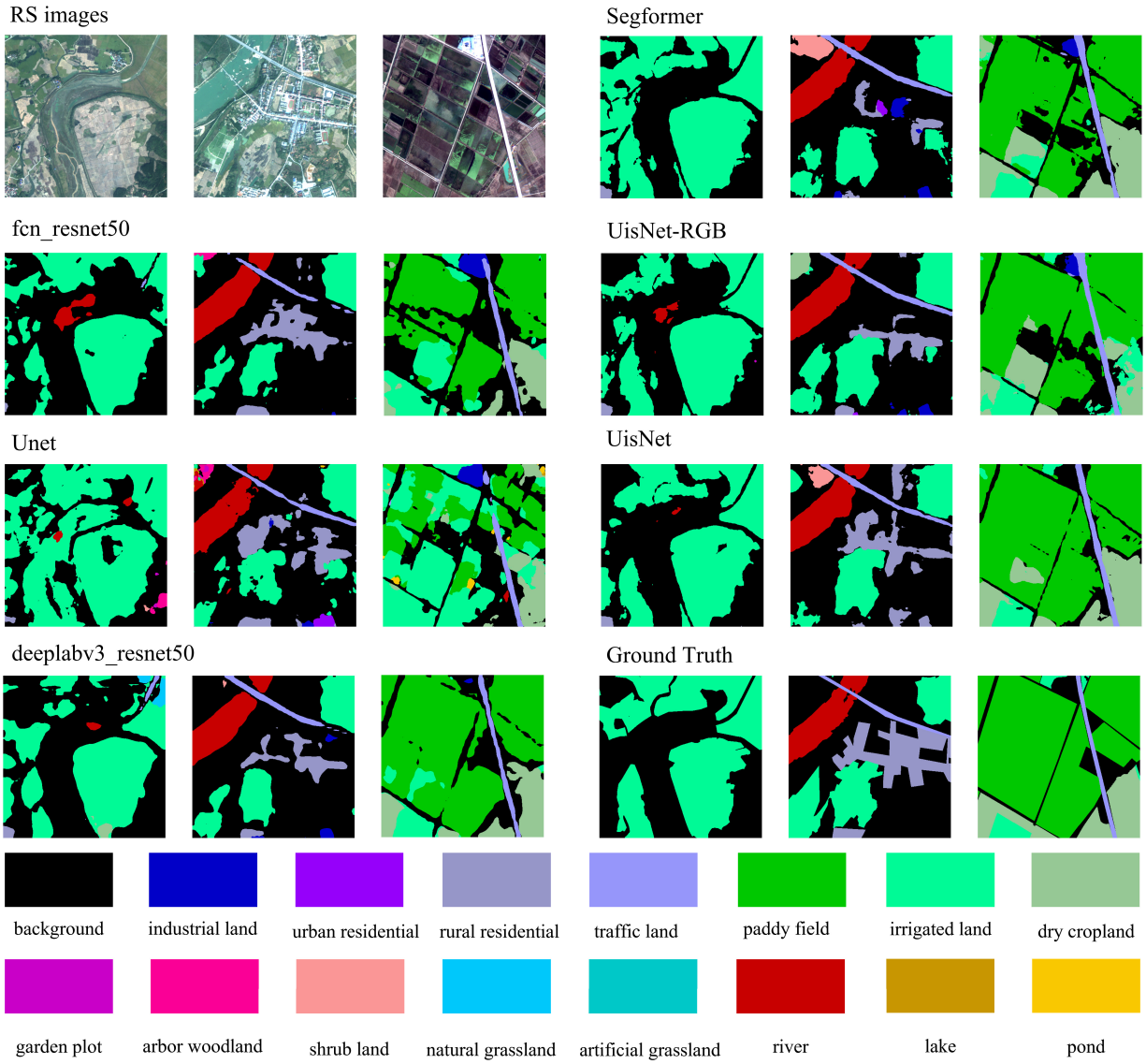

Fig. 14. Illustrations of some semantic segmentation results on GID.

are divided into 5 categories of which UIS areas proportion are within the range of [0, 0.2], (0.2, 0.4], (0.4, 0.6], (0.6, 0.8] or (0.8, 1]. By comparing the mIoU and UIS area proportion in the testing and validation set, the relation between the semantic segmentation uncertainty and the UIS area proportion at the parcel scale can be studied.

In Fig. 13, the boxplots of the mIoU can reflect the mIoU distribution characteristics under different proportions of UIS areas at the parcel scale. Specifically, the box height in the boxplots indicates the interquartile range, and the upper and bottom lines represent the maximum and minimum values in the mIoU. The red line represents the median, and the green line represents the mean value. It can be observed from Fig. 13 that as the UIS area proportion increases, the median and mean value of mIoU become larger, and the distance between the upper and lower lines of the mIoU box becomes smaller. This shows that as the proportion of UIS areas increases, the performance of UIS semantic segmentation improves, and the accuracy uncertainty of the proposed model decreases.

E. Experiments and Comparison on the GID Dataset

To prove the generalization of the proposed UisNet, we conduct experiments on a widely used public dataset, the GID dataset. GID is a large-scale dataset that contains wide distribution and high spatial resolution Gaofen-2 (GF-2) images. GID consists of a large-scale classification set and a fine land cover classification set. In this experiment, the fine land cover classification set with ten pixel-level annotated GF-2 images (with the size of 7200 × 6800) in GID were chosen as the experimental dataset. The dataset is first clipped into nonoverlaped 1820 patches with the size of 512 × 512. Then these patches were divided into a training set with 1092 samples (60% training ratio), a validation set with 364 samples (20% validation ratio), and a testing set with 364 samples (20% testing ratio).

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To prove the effectiveness of the proposed UisNet, we designed multiple comparative experiments using several semantic segmentation models, including FCN, UNet, Deeplab-V3, and SegFormer, as well as UisNet-RGB (UisNet using only RGB bands in GID) and UisNet (UisNet using both RGB and near-infrared (NIR) bands). Table IV shows the IoU (%) per category in GID and mIoU (%) of different models in the testing set.

Of the compared models, the UNet had the least significant mIoU (%) of 52.65. The FCN had the second least significant mIoU (%) of 57.68. The DeepLab-V3 and Segformer achieved similar mIoU (%) of about 58 to 59. However, considering that the number of parameters amount for SegFormer was much less than DeepLab-V3, it is evident that SegFormer can efficiently utilize the model parameters and obtain similar results with fewer parameters. The proposed UisNet-RGB had the second-best mIoU among all the compared models, with an improvement of 0.5% in mIoU over the Segformer model. This indicates the spatial-channel feature fusing module greatly enhanced the semantic segmentation results. The proposed UisNet achieved the most significant mIoU (%) of 60.23 by adding the NIR bands, with an improvement of 1.15% in mIoU. It indicates the proposed model can greatly improve semantic segmentation tasks by effectively fusing multimodality data.

We also provide some semantic segmentation results of the comparison models in the testing set in Fig. 14. It can be observed that our results maintain the geometric characteristics of the objects in a better way, since the segmentation results of our proposed models have flatter edges and more regular shapes than others. For example, the results of FCN and Unet contain large number of internal holes and misclassified salt and pepper noises. The results of SegFormer and UisNet-RGB maintain the geometric characteristics of the objects but contain lots of misclassified parts. The DeepLab-V3 is difficult to accurately segment the geometric characteristics of objects, especially the edges of the objects. Only the proposed UisNet model can accurately segment the objects and maintain the geometric characteristics. It demonstrates the superiority of the proposed UisNet.

IV. Conclusion

In this article, a UIS semantic segmentation method, namely UisNet, utilized a transformer-based block to separately model multimodal data, including HRRS images and building polygon data, and a spatial-channel feature fusing module to fuse the features from multimodal data (pixel-parcel scale features of HRRS images and object scale features of building polygon data) to improve UIS semantic segmentation.

To assess the proposed model's performance, we compared it with several semantic segmentation models. The experiments were conducted in Shenzhen City, and the results confirmed that, by utilizing multimodal data with pixel-parcel and object scale features, the proposed UisNet performed best. This finding proves the superiority of the proposed UisNet with multimodality data in handling UIS mapping tasks. Furthermore, we calculated the proportion of UIS area at a parcel scale to further characterize the UISs' spatial distribution in Shenzhen City. Besides, we add a set of experiments on the GID dataset and compare our method with the current state-of-the-art semantic segmentation methods. Experiments show that the proposed UisNet improves mIoU by 1.64% to 7.58% compared to other methods. This proves the superiority of the proposed UisNet. Our method demonstrates the potential and value of deep learning technology and multimodal data in UISs' semantic segmentation and analysis. However, several countries with large informal settlement areas may lack large-scale building polygon data. In this case, the proposed UisNet can be utilized with a decent quality solution using only HRRS remote sensing images.

References

[1] N. Mboga, C. Persello, J. R. Bergado, and A. Stein, "Detection of informal settlements from VHR images using convolutional neural networks," *Remote Sens.*, vol. 9, no. 11, p. 1106, 2017.

[2] T. K.-A. Williams, T. Wei, and X. Zhu, "Mapping urban slum settlements using very high-resolution imagery and land boundary data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 166–177, 2020.

[3] P. Hofmann, H. Taubenböck, and C. Werthmann, "Monitoring and modelling of informal settlements—A review on recent developments and challenges," in *Proc. Joint Urban Remote Sens. Event (JURSE)*, Apr. 2015, pp. 1–4.

[4] N. J. Yuan, Y. Zheng, X. Xie, Y. Wang, K. Zheng, and H. Xiong, "Discovering urban functional zones using latent activity trajectories," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 3, pp. 712–725, Mar. 2015.

[5] S. Du, S. Du, B. Liu, and X. Zhang, "Mapping large-scale and fine-grained urban functional zones from VHR images using a multi-scale semantic segmentation network and object based approach," *Remote Sens. Environ.*, vol. 261, Aug. 2021, Art. no. 112480.

[6] X. Zhang, S. Du, and Z. Zheng, "Heuristic sample learning for complex urban scenes: Application to urban functional-zone mapping with VHR images and POI data," *ISPRS J. Photogramm. Remote Sens.*, vol. 161, pp. 1–12, Mar. 2020.

[7] R. Fan, R. Feng, W. Han, and L. Wang, "Urban functional zone mapping with a bibranch neural network via fusing remote sensing and social sensing data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 11737–11749, 2021.

[8] G. Cheng, J. Han, and X. Lu, "Remote sensing image scene classification: Benchmark and state of the art," *Proc. IEEE*, vol. 105, no. 10, pp. 1865–1883, Oct. 2017.

[9] G.-S. Xia *et al.*, "AID: A benchmark data set for performance evaluation of aerial scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3965–3981, Jul. 2017.

[10] X. Lu, X. Zheng, and Y. Yuan, "Remote sensing scene classification by unsupervised representation learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 9, pp. 5148–5157, Sep. 2017.

[11] R. Fan, R. Feng, L. Wang, J. Yan, and X. Zhang, "Semi-MCNN: A semisupervised multi-CNN ensemble learning method for urban land cover classification using submeter HRRS images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4973–4987, 2020.

[12] P. S. Thenkabail, M. Schull, and H. Turral, "Ganges and Indus river basin land use/land cover (LULC) and irrigated area mapping using continuous streams of MODIS data," *Remote Sens. Environ.*, vol. 95, no. 3, pp. 317–341, Apr. 2005.

[13] H. Shafizadeh-Moghadam, M. Khazaei, S. K. Alavipanah, and Q. Weng, "Google Earth engine for large-scale land use and land cover mapping: An object-based classification approach using spectral, textural and topographical factors," *GISci. Remote Sens.*, vol. 58, no. 6, pp. 914–928, Aug. 2021.

[14] G. Cheng and J. Han, "A survey on object detection in optical remote sensing images," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 11–28, Jul. 2016.

[15] K. Li, G. Wan, G. Cheng, L. Meng, and J. Han, "Object detection in optical remote sensing images: A survey and a new benchmark," *ISPRS J. Photogramm. Remote Sens.*, vol. 159, pp. 296–307, Jan. 2020.

[16] J. Han, D. Zhang, G. Cheng, L. Guo, and J. Ren, "Object detection in optical remote sensing images based on weakly supervised learning and high-level feature learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 6, pp. 3325–3337, Jun. 2015.

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[17] A. Fallatah, S. Jones, D. Mitchell, and D. Kohli, "Mapping informal settlement indicators using object-oriented analysis in the middle east," *Int. J. Digit. Earth*, vol. 12, no. 7, pp. 802–824, Jul. 2019.

[18] P. Hofmann *et al.*, "Detecting informal settlements from IKONOS image data using methods of object oriented image analysis-an example from Cape Town (South Africa)," in *Remote Sensing of Urban Areas/Fernerkundung in Urbanen Räumen*, C. Jürgens, Ed. 2001, pp. 41–42.

[19] A. Fallatah, S. Jones, L. Wallace, and D. Mitchell, "Combining object-based machine learning with long-term time-series analysis for informal settlement identification," *Remote Sens.*, vol. 14, no. 5, p. 1226, Mar. 2022.

[20] J. Duque, J. Patino, and A. Betancourt, "Exploring the potential of machine learning for automatic slum identification from VHR imagery," *Remote Sens.*, vol. 9, no. 9, p. 895, Aug. 2017.

[21] M. Kuffer, K. Pfeffer, R. Sliuzas, and I. Baud, "Extraction of slum areas from VHR imagery using GLCM variance," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 9, no. 5, pp. 1830–1840, May 2016.

[22] C. Persello and A. Stein, "Deep fully convolutional networks for the detection of informal settlements in VHR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2325–2329, Dec. 2017.

[23] M. Wurm, T. Stark, X. X. Zhu, M. Weigand, and H. Taubenböck, "Semantic segmentation of slums in satellite images using transfer learning on fully convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 150, pp. 59–69, Apr. 2019.

[24] S. Lang, "Object-based image analysis for remote sensing applications: Modeling reality–dealing with complexity," in *Object-Based Image Analysis*. Springer, 2008, pp. 3–27.

[25] J. Pratomo, M. Kuffer, J. Martinez, and D. Kohli, "Coupling uncertainties with accuracy assessment in object-based slum detections, case study: Jakarta, Indonesia," *Remote Sens.*, vol. 9, no. 11, p. 1164, Nov. 2017.

[26] G. Leonita, M. Kuffer, R. Sliuzas, and C. Persello, "Machine learning-based slum mapping in support of slum upgrading programs: The case of Bandung city, Indonesia," *Remote Sens.*, vol. 10, no. 10, p. 1522, Sep. 2018.

[27] M. R. Ibrahim, H. Titheridge, T. Cheng, and J. Haworth, "PredictSLUMS: A new model for identifying and predicting informal settlements and slums in cities from street intersections using machine learning," *Comput., Environ. Urban Syst.*, vol. 76, pp. 31–56, Jul. 2019.

[28] B. J. Gram-Hansen *et al.*, "Mapping informal settlements in developing countries using machine learning and low resolution multi-spectral data," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, 2019, pp. 361–368.

[29] J. A. Benediktsson, J. A. Palmason, and J. R. Sveinsson, "Classification of hyperspectral data from urban areas based on extended morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 480–491, Mar. 2005.

[30] M. Fauvel, J. A. Benediktsson, J. Chanussot, and J. R. Sveinsson, "Spectral and spatial classification of hyperspectral data using SVMs and morphological profiles," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 11, pp. 3804–3814, Nov. 2007.

[31] M. Wurm, H. Taubenböck, M. Weigand, and A. Schmitt, "Slum mapping in polarimetric SAR data using spatial features," *Remote Sens. Environ.*, vol. 194, pp. 190–204, Jun. 2017.

[32] K. K. Owen and D. W. Wong, "An approach to differentiate informal settlements using spectral, texture, geomorphology and road accessibility metrics," *Appl. Geography*, vol. 38, pp. 107–118, Mar. 2013.

[33] A. Fallatah, S. Jones, and D. Mitchell, "Object-based random forest classification for informal settlements identification in the middle east: Jeddah a case study," *Int. J. Remote Sens.*, vol. 41, no. 11, pp. 4421–4445, Jun. 2020.

[34] S. Du, F. Zhang, and X. Zhang, "Semantic classification of urban buildings combining VHR image and GIS data: An improved random forest approach," *ISPRS J. Photogramm. Remote Sens.*, vol. 105, pp. 107–119, Jul. 2015.

[35] X. Huang, H. Liu, and L. Zhang, "Spatiotemporal detection and analysis of urban villages in mega city regions of China using high-resolution remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 7, pp. 3639–3657, Jul. 2015.

[36] A. Ajami, M. Kuffer, C. Persello, and K. Pfeffer, "Identifying a slums' degree of deprivation from VHR images using convolutional neural networks," *Remote Sens.*, vol. 11, no. 11, p. 1282, 2019.

[37] D. Verma, A. Jana, and K. Ramamritham, "Transfer learning approach to map urban slums using high and medium resolution satellite imagery," *Habitat Int.*, vol. 88, Jun. 2019, Art. no. 101981.

[38] R. Prabhu, B. Parvathavarthini, and A. R. Alaguraja, "Integration of deep convolutional neural networks and mathematical morphology-based postclassification framework for urban slum mapping," *J. Appl. Remote Sens.*, vol. 15, no. 1, Mar. 2021, Art. no. 014515.

[39] B. Ayo, "Integrating openstreetmap data and Sentinel-2 imagery for classifying and monitoring informal settlements," Ph.D. dissertation, 2020.

[40] C. Persello and M. Kuffer, "Towards uncovering socio-economic inequalities using VHR satellite images and deep learning," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Sep. 2020, pp. 3747–3750.

[41] J. Mast, C. Wei, and M. Wurm, "Mapping urban villages using fully convolutional neural networks," *Remote Sens. Lett.*, vol. 11, no. 7, pp. 630–639, Jul. 2020.

[42] I. M. Bello, K. Zhang, J. Wang, and H. Li, "Lightweight multiscale framework for segmentation of high-resolution remote sensing imagery," *J. Appl. Remote Sens.*, vol. 15, no. 3, Aug. 2021, Art. no. 034508.

[43] Z. Ye *et al.*, "Mapping and discriminating rural settlements using Gaofen-2 images and a fully convolutional network," *Sensors*, vol. 20, no. 21, p. 6062, Oct. 2020.

[44] M. Dixit, K. Chaurasia, and V. K. Mishra, "Automatic building extraction from high-resolution satellite images using deep learning techniques," in *Proc. Int. Conf. Paradigms Comput., Commun. Data Sci.* Springer, 2021, pp. 773–783.

[45] L. C. Chen, G. Papandreou, and I. Kokkinos, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Jun. 2017.

[46] Y. Wang *et al.*, "Mask DeepLab: End-to-end image segmentation for change detection in high-resolution remote sensing images," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 104, Dec. 2021, Art. no. 102582.

[47] Y. Li, W. Xu, H. Chen, J. Jiang, and X. Li, "A novel framework based on mask R-CNN and histogram thresholding for scalable segmentation of new and old rural buildings," *Remote Sens.*, vol. 13, no. 6, p. 1070, Mar. 2021.

[48] I. Kotaridis and M. Lazaridou, "Remote sensing image segmentation advances: A meta-analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 173, pp. 309–322, Mar. 2021.

[49] C. Zhang *et al.*, "An object-based convolutional neural network (OCNN) for urban land use classification," *Remote Sens. Environ.*, vol. 216, pp. 57–70, Oct. 2018.

[50] W. Zhou, D. Ming, X. Lv, K. Zhou, H. Bao, and Z. Hong, "SO–CNN based urban functional zone fine division with VHR remote sensing image," *Remote Sens. Environ.*, vol. 236, Jan. 2020, Art. no. 111458.

[51] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," 2021, *arXiv:2105.15203*.

[52] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[53] A. Dosovitskiy *et al.*, "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. 9th Int. Conf. Learn. Represent. (ICLR)*, May 2021, pp. 1–22.

[54] A. Srinivas, T.-Y. Lin, N. Parmar, J. Shlens, P. Abbeel, and A. Vaswani, "Bottleneck transformers for visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2021, pp. 16519–16529.

[55] A. Vaswani *et al.*, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 5998–6008.

[56] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[57] D. Hong *et al.*, "SpectralFormer: Rethinking hyperspectral image classification with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2022.

[58] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[59] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2014.

[60] L. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. 15th Eur. Conf. Comput. Vis.* (Lecture Notes in Computer Science), vol. 11211, V. Ferrari, M. Hebert, C. Sminchisescu, and Y. Weiss, Eds. Munich, Germany: Springer, Sep. 2018, pp. 833–851.

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[61] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Long Beach, CA, USA, Jun. 2019, pp. 3146–3154.
[62] W. Lu, C. Tao, H. Li, J. Qi, and Y. Li, "A unified deep learning framework for urban functional zone extraction based on multi-source heterogeneous data," *Remote Sens. Environ.*, vol. 270, Mar. 2022, Art. no. 112830.
[63] J. Li *et al.*, "Deep learning in multimodal remote sensing data fusion: A comprehensive review," 2022, *arXiv:2205.01380*.
[64] R. Fan, J. Li, W. Song, W. Han, J. Yan, and L. Wang, "Urban informal settlements classification via a transformer-based spatial–temporal fusion network using multimodal remote sensing and time-series human activity data," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 111, Jul. 2022, Art. no. 102831.
[65] J. Yin *et al.*, "Integrating remote sensing and geospatial big data for urban land use mapping: A review," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 103, Dec. 2021, Art. no. 102514.
[66] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3rd Int. Conf. Learn. Represent. (ICLR)*, Y. Bengio and Y. LeCun, Eds. San Diego, CA, USA, May 2015.
[67] S. Jadon, "A survey of loss functions for semantic segmentation," in *Proc. IEEE Conf. Comput. Intell. Bioinf. Comput. Biol. (CIBCB)*, Oct. 2020, pp. 1–7.
[68] M. Cordts *et al.*, "The cityscapes dataset for semantic urban scene understanding," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 3213–3223.

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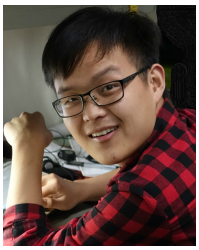

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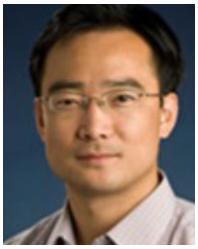

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